

CHAPTER ONE

Introduction

1.1 Background of the Study

Pain management remains a fundamental component of medical practice, with opioids playing a significant role in the treatment of moderate to severe pain. Over the years, opioids have been widely used in clinical settings due to their potent analgesic properties and ability to improve patient comfort and functional capacity. These drugs are especially valuable in the management of acute pain, chronic conditions, cancer-related pain, and palliative care, where alternative therapies may be inadequate. (Hyland et al., 2021)

Despite their undeniable therapeutic benefits, the use of opioids has been accompanied by growing concerns regarding misuse, dependence, and addiction. Prolonged or inappropriate opioid use has been linked to significant health risks, including respiratory depression, overdose, and opioid use disorder. The dual nature of opioids—as effective pain relievers and substances with high abuse potential—has generated global debate regarding their role in modern medicine. (Paul *et al.*, 2021) In recent years, opioid misuse has emerged as a major public health challenge, extending beyond high-income countries to developing nations. In Nigeria, the increasing availability and misuse of opioids such as tramadol and codeine have contributed to rising rates of addiction, particularly among young and vulnerable populations. Weak regulatory frameworks, easy access to opioids, and limited addiction treatment services have further worsened the situation. (Al Hassan, 2022)

This seminar explores the complex balance between the therapeutic use of opioids for pain relief and their potential to cause addiction. By examining the pharmacological properties of opioids, their clinical applications, and the mechanisms and consequences of opioid addiction, this study aims to provide a comprehensive understanding of opioids as both beneficial and potentially harmful agents in healthcare.

1.2 Definition of Terms

Opioids: A class of drugs that act on opioid receptors in the nervous system to produce analgesia, sedation, and euphoria.

Opioid Receptors: Specific receptors (mu, kappa, and delta) in the central and peripheral nervous systems that mediate the effects of opioids.

Analgesia: The relief or reduction of pain without loss of consciousness.

Tolerance: A state in which increasing doses of a drug are required to achieve the same therapeutic effect.

Dependence: A physiological adaptation to a drug characterized by withdrawal symptoms upon cessation.

Addiction: A chronic disorder marked by compulsive drug use despite harmful consequences.

Opioid Use Disorder: A medical condition characterized by problematic opioid use leading to significant impairment or distress.

Palliative Care: Specialized medical care focused on providing relief from pain and other distressing symptoms in patients with serious illnesses.

1.3 Methodologies

This seminar adopts a narrative review approach, utilizing secondary data obtained from peer-reviewed journal articles, review papers, and public health publications related to opioids. Relevant literature was sourced from reputable medical and scientific databases to ensure accuracy and reliability. The selected articles provided information on opioid pharmacology, clinical use in pain management, mechanisms of addiction, and the public health impact of opioid misuse.

The collected literature was critically analyzed and synthesized to present a balanced discussion of opioids as both therapeutic agents and substances with addiction potential. Emphasis was placed on integrating pharmacological principles with clinical and public health perspectives to support informed conclusions and recommendations.